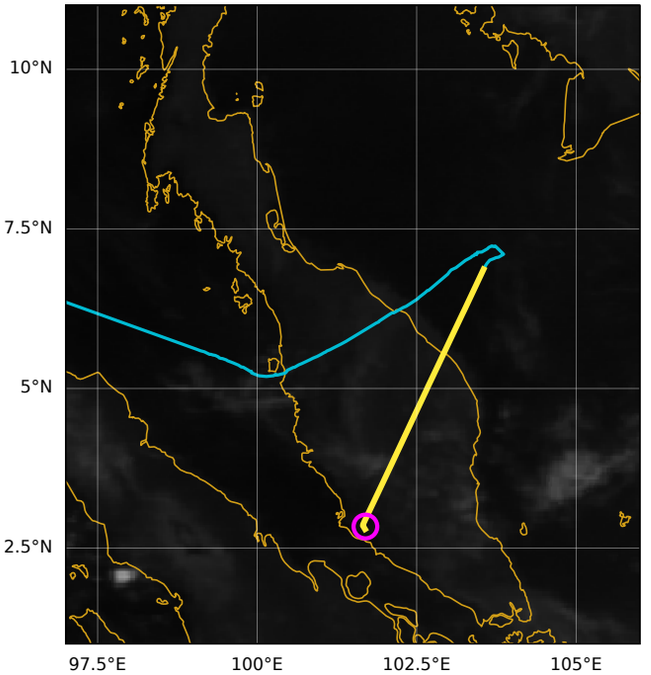
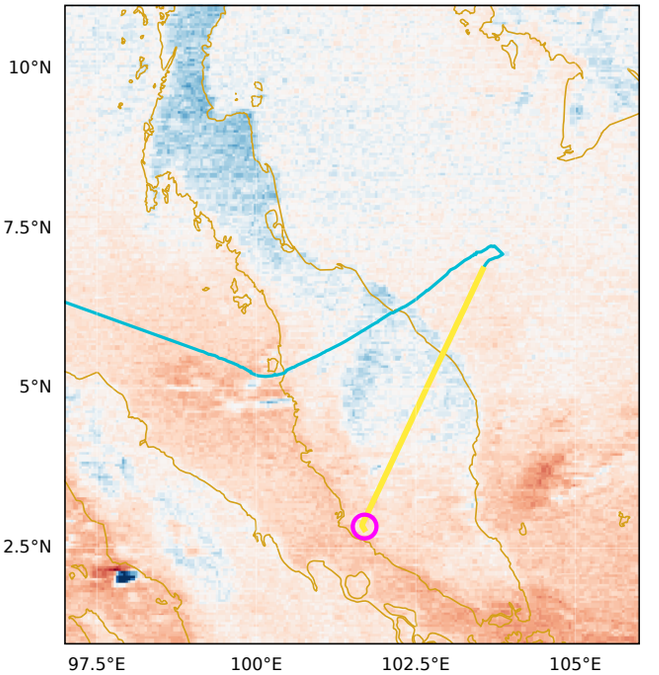
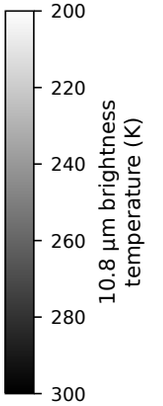
Control check: archived infrared images along the known radar track, 7 March 2014

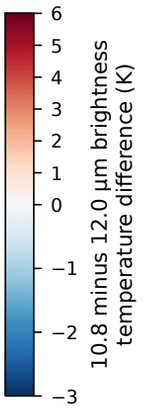
FY-2D full disc, scan start 16:31:00
10.8 µm brightness temperature; aircraft reached 16:43 UTC



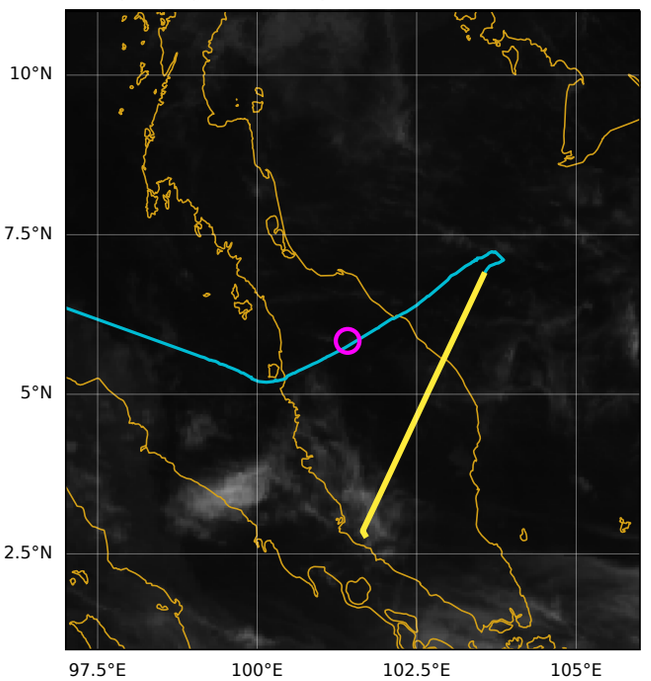
FY-2D full disc, scan start 16:31:00
10.8 minus 12.0 µm difference; aircraft reached 16:43 UTC



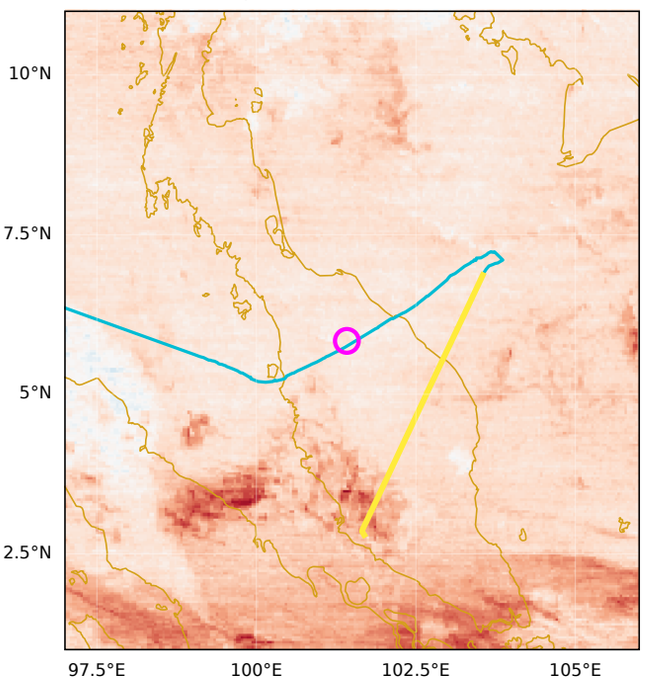




MTSAT-2 full disc, scan start 17:31:15
10.8 µm brightness temperature; aircraft reached 17:43 UTC



MTSAT-2 full disc, scan start 17:31:15
10.8 minus 12.0 µm difference; aircraft reached 17:43 UTC



Military radar track
16:42-18:22 UTC

Segment 16:42-17:20 UTC
(ice-supersaturated air at
FL390-FL400 near the track)

Aircraft apparent position
when the scan reached it

Coastline

Plain note: This is a control check, not a search. It shows the two archived images taken while the aircraft was still on the military radar track near the Malay Peninsula. Weather analysis shows ice-supersaturated air at 39,000-40,000 ft near this part of the track, where condensation trails of other aircraft could persist. If thin straight lines or thin ice cloud show here, the images can show such features; if not, they set the limit of what these images can show.

Technical note: FY-2D 86.45 E (CEReS re-encoded HRIT, 5 km nadir, scan start = archive label, N->S 25 min modelled) and MTSAT-2 145.0 E (CEReS gridded 0.04 deg, line times from the HRIT header). Radar track: Geoscience Australia digital military radar data (Data-Cleaned-Estim-Time); apparent position = line of sight through the aircraft at FL350 meets the surface. Ice supersaturation at FL390-FL400 between Kuala Lumpur and IGARI and too-warm air at FL350 on 16:42-17:20: ERA5, parallel contrail-physics study (contrail-notes.md, artifact df5975a4). Split-window difference is positive for thin ice cloud and also for clear moist tropical air. Visual check only; no detection algorithm.